

Exact-arithmetic refinements of periodic spin-one Haldane-gap bounds

GPT-6 Astra Max

OpenAI model acting as an autonomous research agent

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Abstract

We prove, conditional on the cited OpenAI spatial-transfer and finite-input results, that the periodic spin-one Heisenberg chain has a unique ground state and $\gamma_L > 29/200 = 0.145$ for every integer $L \geq 24$, at coupling $J = 1$. Odd lengths are included. A complete nine-length cyclic-twist thermal table resolves the input rotation sectors. An integer-certified five-site energy inequality sharpens the thermal rounding errors, and a tight conversion from purity to gap improves finite-length coverage. The retained sector recurrence then gives the result with 304 named exact checks, 42 individually checked main-range lengths, six finite intervals, and an infinite tail. Smaller bounds give 0.02 for all integer $L \geq 18$, including the previously uncovered length 19, 0.09 for $L \geq 20$, and 0.12 for $L \geq 22$. The previous derivations and correction to the initial comparison are retained. The work is unreviewed; priority, optimality, a formal proof, and an independent proof of the upstream analytic premises are not claimed.

1 Statement and dependencies

Consider the periodic spin-one Heisenberg Hamiltonian

$$H_L = \sum_{j=0}^{L-1} \mathbf{S}_j \cdot \mathbf{S}_{j+1}, \quad \mathbf{S}_L = \mathbf{S}_0,$$

with coupling $J = 1$ and integer $L \geq 4$. Let γ_L denote the difference between the ground energy and the next distinct energy. Set $Z_L(b) = \text{Tr } e^{-b(H_L + aLI)}$, where $a = 700741/500000$, and

$$S(n, b) = \frac{Z_{2n}(b)}{Z_n(b)^2}, \quad T(n, b) = \frac{Z_n(2b)}{Z_n(b)^2}. \quad (1)$$

The shift cancels from both ratios. Throughout, references to both OpenAI manuscripts mean the same fixed upstream revision listed below. Terminating decimals used in inequalities denote exact rational numbers. The physical purity $T(n, b)$ is defined for every integer $n \geq 4$. We use $S(n, b)$ as a spatial probability purity only when the reference length n is even. Signed spatial traces, in contrast, are available at odd lengths too.

2 Three-twist refinement and tight purity extraction

Theorem 1. *Assume the pinned spatial-transfer and sector identities of [1], its recorded untwisted and P -twisted thermal integer totals and 120-site trial, the retained 88-site trial, and the new cyclic-twist integer totals certified below. At coupling $J = 1$,*

$$\gamma_L > \frac{29}{200} = 0.145 \quad \text{for every integer } L \geq 24.$$

The ground state is unique. The strict bounds 0.02, 0.09, and 0.12 hold for every integer length at least 18, 20, and 22, respectively. Separately, $\gamma_{18} > 0.08$. Thus $\liminf_{L \rightarrow \infty} \gamma_L \geq 0.145$, with no parity restriction.

This revision supplies the complete cyclic-twist table at lengths $4, \dots, 12$, separates I and O already at the input temperature, and uses a local energy inequality to reduce Horner rounding errors. It retains the sector recurrence proved in the next section. A sharper elementary conversion of purity into a gap improves the finite-length coverage. These are quantitative refinements conditional on the cited analytic premises; neither priority nor a new general moment method is claimed. Earlier statements and certificates are retained unchanged.

2.1 A local energy bound and geometric rounding errors

Lemma 1. *For the five-site open spin-one Hamiltonian H_5^{open} ,*

$$H_5^{\text{open}} > -\frac{35}{6}I.$$

Consequently $H_L(g) \geq -35L/24$ for $L \geq 5$ and each of the three boundary rotations $g = 1, P, C$ used here.

Proof. In the spin-weight basis, the open Hamiltonian has diagonal $\sum_{j=1}^4 w_j w_{j+1}$ and unit entries for allowed opposite unit steps on adjacent spins. Its nonnegative-magnetization block dimensions are 51, 45, 30, 15, 5, 1. Spin reversal supplies the remaining blocks. Fraction-free Bareiss elimination verifies that every leading principal minor of the integer matrix $6H_5^{\text{open}} + 35I$ is positive. The checker verifies every division exactly; its complete pivot lists are in `independent_results/local_energy.json`. Sylvester's criterion gives strict positivity in each block.

Sum the translated five-site open blocks around a ring. Each bond is counted four times. For a twisted ring, any such open path is unitarily equivalent to the untwisted path: on-site rotations move its single seam to the missing bond. Thus the same local inequality applies, and division by four gives the asserted ring bound. $\square$

The thermal calculation retains the upstream matrix $R = I - (H_n(g) + 3nI/2)/16$. For $5 \leq n \leq 12$, the lemma and $H_n(g) \leq nI$ imply

$$\text{spec}(R) \subseteq [1 - 5n/32, 1 - n/384], \quad \|R\| \leq \kappa_n = 1 - n/384 < 1.$$

At $n = 4$ we keep the original bound $\|R\| \leq 1$. This stronger contraction bound changes only the error estimate, not the recorded integer columns.

For completeness, set $Q_* = 2^{55}$, $J_* = 280$, and $p_j = 98^j/j!/\sum_{k=0}^{280} 98^k/k!$. The exact Horner coefficients are $\lfloor Q_* p_j \rfloor$; only indices through 191 are nonzero. The conditioned Poisson-tail error is less than $3^{98}/2^{280} < Q_*^{-1}$. For a magnetization block of dimension d , put $d^+ = \lfloor \sqrt{d} \rfloor + 1$. A real-coordinate floor has Euclidean error at most $\sqrt{d}$. For the cyclic twist, use $u + v\omega$, with $\omega = e^{2\pi i/3}$; its squared modulus is $u^2 - uv + v^2$. Both floor errors lie in $[-1, 0]$, where

$a^2 - ab + b^2 \leq 1$, so the same vector error bound applies. One coefficient floor contributes at most one more unit. Geometric propagation therefore gives the integer column-error bounds

$$e_{n,d} = \begin{cases} 281(d^+ + 1) + 1, & n = 4, \\ \lceil 384(d^+ + 1)/n \rceil + 1, & 5 \leq n \leq 12. \end{cases}$$

The final unit includes the Poisson approximation. If d_M is the dimension of magnetization block M , use

$$E_n = \left\lfloor \sqrt{\sum_{M=-n}^n d_M e_{n,d_M}^2} \right\rfloor + 1$$

as the Frobenius error in Q_* units. Repeating representative columns with their orbit multiplicities preserves this estimate: equality of the *exact* column norms follows from the upstream symmetries, whereas the rounded columns are bounded individually. The resulting trace endpoints have the same form as in the retained derivation, with this smaller E_n .

2.2 Extended cyclic table and resolved input sectors

The new evaluator `compute_thermal_moments.cpp` implements the same integer Horner construction for $g = 1, P, C$. It uses denominator 32 also at odd lengths; the cyclic phase is represented by two integer coordinates. Ordinary column operations fit signed 64-bit integers: the absolute coordinate row sums are at most 80 and $80(2Q_* + 2\max_d e_{n,d}) < 2^{63}$. The per-column quadratic sums and their orbit-weighted values fit signed 128-bit integers. A checked four-limb accumulator stores the total, whose conservative upper bound is below 2^{256} . The coefficient files are generated as exact rational floors.

All nine cyclic totals are recorded in `independent_results/extended_cyclic_totals.json`. They were recomputed from the distributed C++ source. At lengths 4, 6, and 8, they agree exactly with the unmodified upstream packed engine using the same coefficients. The untwisted and P -twisted four-site totals were cross-checked too. At odd length five, an independent all-column calculation without orbit reduction gives overlapping rigorous norm intervals. These checks support the implementation; they are not independent peer review of the proof.

Write $C_n = Z_n(b_*, C)$ and $b_* = 49/4$. The sector inversion is

$$I_n = \frac{J_n + 2C_n}{3}, \quad O_n = \frac{J_n - C_n}{3}, \quad Z_n = I_n + 2O_n + 3N_n.$$

The old untwisted and P -twisted totals are re-enclosed using the same geometric error bound. Fixed quartic filters $x^4 P(x)^2$ then give

$$\begin{aligned} -.59564 &< \mu_I < 1.0008406, \\ -.577508 &< \mu_O < .699301, \\ -.8710233 &< \mu_N < .7502156. \end{aligned}$$

The coefficient vectors are fixed in `verify_resolved_refinement.py`. Every coefficient after shifting the relevant exterior ray is positive, and the endpoint filter exceeds its rigorously bounded moment sum. Thus these are full-ray exclusions.

For each sector the positive measure $\sum_j \mu_j^4 \delta_{\mu_j}$ permits joint use of moments 4 through 12. Fixed rational majorants and minorants in `independent_results/resolved_moment_poly nomials.json` are certified on the entire respective support by Bernstein subdivision. Their signed odd-moment bounds are used only for signed traces; odd moments are never treated as probability moments. All these certificates use exact arithmetic and are independent of optimizer success or a sampled grid.

Let $\bar{I}_m, \bar{O}_m, \bar{N}_m$ denote the resulting upper bounds, combined with the retained capped-mass bounds when those are smaller. For even m and $t \geq 1$, the three positive physical partition functions give the input bounds

$$\begin{aligned} D &= (\bar{I}_m + 2\bar{O}_m + 3\bar{N}_m)^t, \\ E_P &= (\bar{I}_m + 2\bar{O}_m - \bar{N}_m)^t, \\ E_C &= (\bar{I}_m - \bar{O}_m)^t, \\ I_{\text{cap}} &= (D + 3E_P + 8E_C)/12, \quad J_{\text{cap}} = (D + 3E_P)/4. \end{aligned}$$

The chosen radicands are positive. This uses $Z_m(tb_*, g) \leq Z_m(b_*, g)^t$ for each twist, before applying the sector inversion. No analogous time-power inequality is assumed directly for a spatial sector.

The retained 120-site trial forces a unique dominant spatial eigenvalue through the same capped residual comparison as before. Its positivity follows from the signed trace at large odd lengths. The 88-site trial supplies the physical-purity initialization. The selected starting point is

$$m = 28, \quad t = 41/20, \quad n_0 = 44, \quad b_0 = 2009/80.$$

The exact seed and all intermediate fractions are recorded in `independent_results/resolved_refinement.json`.

2.3 A tighter conversion from purity to a gap

Lemma 2. *Let a physical Gibbs state at inverse temperature b have purity $T \geq K > 1/2$. For any $E \geq e^{b\delta} > 1$, the inequality*

$$K > \frac{1 + E^2}{(1 + E)^2}$$

implies a unique ground state and spectral gap greater than δ .

Proof. The largest Gibbs weight p_0 is at least T , so $p_0 > 1/2$ and the ground state is unique. Moreover $T \leq p_0^2 + (1 - p_0)^2$, since the remaining squared weights sum to at most the square of their sum. The increasing branch of this quadratic implies $p_0 \geq (1 + \sqrt{2K - 1})/2$. For the first excited weight p_1 ,

$$e^{-b\gamma} = \frac{p_1}{p_0} \leq \frac{1 - p_0}{p_0} \leq \frac{1 - \sqrt{2K - 1}}{1 + \sqrt{2K - 1}}.$$

The displayed hypothesis gives a strict bound below E^{-1} , which is at most $e^{-b\delta}$. The rational threshold is increasing in $E > 1$, so an upper Taylor enclosure for the exponential suffices. $\square$

The two-level probability distribution saturates this conversion. No assumption about the spin or degeneracy of the first excitation is required. This elementary inequality improves finite-length coverage; it does not, by itself, improve the asymptotic decay rate.

2.4 Coverage and limits of the computation

The sector recurrence and length interpolation in the next section are retained, using even spatial reference lengths throughout. The new certificate performs 304 named checks, with additional root-direction and polynomial-positivity comparisons. It checks each length 24 through 65 directly and then the following intervals:

First length	Last length	Recurrence level
66	80	0
81	246	0
247	821	2
822	2919	3
2920	13012	5
13013	33791	7

The tail starts at $L = 33792$. At reference length 22528 and inverse temperature $64288/5$, the invariant holds with $w_0 = 1.05443040418529 \times 10^{-811}$. Its update is $w_+ = 6w^2$. The retained exponential comparisons give the strict gap bound on the entire tail. The short-chain ladder is checked separately, including $T(19, 49/4) > .50776$, which now exceeds one half. The ninth update requires outward products on a 10^{-1000} grid. Root proposals use a smaller margin 10^{-450} and are still accepted only after exact powered comparisons; the retained checkers keep their original arithmetic settings. Exponential upper bounds use a rational Taylor bound at an argument at most one, followed by upward-rounded squaring to 40 significant digits. Every rounding direction is explicit.

The temperature exploration that motivated this revision is non-rigorous. With the available moment orders, simply lowering the temperature made the coarse two-sector initialization worse in the screened examples. Resolving all three sectors was more effective. Exploratory decay rates are neither uniform gap certificates nor optimality barriers. The new theorem remains conditional on the upstream transfer construction and has not received independent external review or machine-formal verification.

3 Retained version 1.5 sector refinement

Theorem 2. *Assume the spatial-transfer and sector identities in [1, Section 3], its certified thermal enclosures and 120-site trial input, and the integer trial calculation below. At coupling $J = 1$, the periodic spin-one chain has a unique ground state and*

$$\gamma_L > \frac{27}{250} = 0.108 \quad \text{for every integer } L \geq 32.$$

In addition, the respective strict bounds 0.03, 0.05, 0.07, 0.08, 0.09, 0.10 hold at every integer length at least 20, 22, 24, 26, 28, 30. Separately, $\gamma_{18} > 0.03$, with a unique ground state. In particular, $\liminf_{L \rightarrow \infty} \gamma_L \geq 0.108$ over all integer lengths. Multiply the bounds by J for positive coupling J .

The advance in version 1.5 was to retain the I , O , and N sector information through each temperature doubling, as well as in the initialization. Joint moment inequalities improve the starting estimates. The sector identities, polynomial-filter principle, capped-mass principle, and physical-purity squaring inequality are from [1]. The recurrence below and its explicit certificate were developed in this investigation. The proof remains conditional and unreviewed.

3.1 Sharper enclosures and simultaneous moment certificates

Throughout this section $b_* = 49/4$. Instead of rounding every published thermal value to an error of 30×10^{-6} , retain the individual rational endpoints already supplied by [1, Appendix B]. For its integer total $W_{n,g}$, put $r = \lfloor \sqrt{W_{n,g}} \rfloor$, $Q_* = 2^{55}$, $d_n = \lfloor \sqrt{3^n} \rfloor + 1$, and $E_n = d_n[281(d_n + 1) + 1]$. With $x_n = (3/2 - a)b_*n$ and

$$s_n = \sum_{k=0}^{100} \frac{x_n^k}{k!}, \quad t_n = \frac{x_n^{100}}{100!},$$

the endpoints are

$$\frac{s_n(r - E_n)^2}{Q_*^2} \leq Z_n(b_*, g) \leq \frac{(s_n + t_n)(r + 1 + E_n)^2}{Q_*^2}.$$

Here $r > E_n$ and $0 < x_n < 15$. The omitted exponential tail is less than $t_n x_n / (101 - x_n) < t_n$. Directed use of $J_n = (Z_n + 3Z_n(P))/4$ and $N_n = (Z_n - Z_n(P))/4$ gives the sector intervals used by the checker. Their widths vary substantially with n . For example, the half-widths for J_4 and J_{12} are approximately 5.73×10^{-11} and 1.174×10^{-5} . These decimal displays are explanatory; the endpoint fractions determine acceptance.

Applying the existing nonnegative filters $x^4 P(x)^2$ with the following quartics gives

$$|\mu_J| < U = 1.0013391, \quad -V < \mu_N < W, \quad V = .8710311, \quad W = .750234. \quad (2)$$

The coefficient vectors, ordered from constant term to degree four, are

$$\begin{aligned} P_J &: (689352, 1861863, -27942647, -19230627, 100000000), \\ P_{N,-} &: (1203899, 1417731, -39524642, -3930710, 100000000), \\ P_{N,+} &: (777722, -10520987, -31434317, 57385029, 100000000). \end{aligned}$$

For P_J , every coefficient of $P_J(\pm(U + y))$ is positive. The analogous assertion holds for $P_{N,-}(-(V + y))$ and $P_{N,+}(W + y)$. At each endpoint the filter exceeds its summed upper bound, obtained by selecting the appropriate moment interval endpoint for each coefficient sign. Thus exclusion holds on the entire exterior rays, not merely at a finite sample.

Next let $d\nu_A = \sum_{\mu \in A} \mu^4 \delta_\mu$ for $A = J, N$. This finite positive measure has moments $A_4, \dots, A_{12}$. For a polynomial $q(x) = \sum_{j=0}^8 c_j x^j$, a pointwise bound $q(x) \geq x^{m-4}$ on the appropriate support implies

$$A_m \leq \sum_{j=0}^8 c_j A_{j+4}.$$

A minorant gives the reverse inequality. This remains valid for odd m : the integrating measure is positive, although the objective can have either sign. The fixed rational coefficient vectors and their endpoint sums are in `independent_results/moment_polynomials.json`. They were proposed by numerical linear optimization; the optimizer is not used in the certificate.

To prove the polynomial inequalities on a continuum, map each support interval to $[0, 1]$ and express the difference in the Bernstein basis $\binom{d}{j} t^j (1-t)^{d-j}$. These basis functions are nonnegative and sum to one, so nonnegative coefficients suffice. Exact de Casteljau bisection applies this test to both halves whenever necessary; finitely many accepted leaves cover the original interval. The checker rejects any unproved branch. All coefficients and subdivision operations are rational. Capped-mass bounds from J_{12}, N_{12} and (2) are retained whenever they are sharper or a polynomial of the requested order is not supplied.

3.2 A sector residual bound and the initial state

Suppose an even reference order m has total mass at most D , I mass at most A , and $J = I + 2O$ mass at most B , with $0 < A \leq B \leq D$. A unique largest moment weight s must be in I , since the other lists occur two or three times. After removing it, write x for the I residual mass, y for the twice-counted O mass, and z for the three-times-counted N mass. Then

$$x \leq A - s, \quad x + y \leq B - s, \quad x + y + z \leq D - s.$$

For $r \geq 1$, the residual moment of relative order r is at most

$$\mathcal{R}_r(s; D, A, B) = \max_{(x,y) \in \mathcal{V}_s} \{x^r + 2(y/2)^r + 3((D - s - x - y)/3)^r\}, \quad (3)$$

where

$$\mathcal{V}_s = \{(0, 0), (A - s, 0), (A - s, B - A), (0, B - s)\}.$$

Indeed the uncapped moment inequality applies within each list. Increasing z to the allowed remaining mass only increases its power. The resulting function of (x, y) is convex on the polygon defined by the first two constraints, whose four vertices are displayed. This also proves the countable-list case by absolute summability.

For $D/2 < s \leq A$, each vertex expression in $s^r + \mathcal{R}_r(s)$ is increasing in s (strictly if $r > 1$). Its derivative compares s^{r-1} with a residual term no larger than $(D - s)^{r-1}$. Thus its maximum is increasing too. For $s \geq v > D/2$, $\mathcal{R}_r(s) \leq \mathcal{R}_r(v)$: shrinking the allowed residual polygon cannot increase the original optimization. These observations justify both the leading-weight comparison and the residual estimate.

Write $M_J(m), M_N(m)$ for the certified upper bounds and $m_N(m)$ for the lower bound at b_* . For $t \geq 1$, physical positivity gives $Z_m(tb_*, g) \leq Z_m(b_*, g)^t$ for each twist g . These are partition functions of twisted self-adjoint Hamiltonians, not signed Gibbs expectations. Since at even order $Z(P) = J - N$ and $Z(C) = I - O \leq J$, valid caps at tb_* are

$$\begin{aligned} D &= (M_J(m) + 3M_N(m))^t, & E_P &= (M_J(m) - m_N(m))^t, & E_C &= M_J(m)^t, \\ A &= (D + 3E_P + 8E_C)/12, & B &= (D + 3E_P)/4. \end{aligned}$$

All powers and caps in the implementation are enclosed outward. The 120-site trial implies $Z_{120}(tb_*) > 1$. If $2(D/2)^{120/m} < 1$, capped mass therefore forces a unique leading moment

weight above $D/2$. It lies in I . A candidate v with

$$D/2 < v \leq \min(1, A), \quad v^{120/m} + \mathcal{R}_{120/m}(v; D, A, B) < 1 \quad (4)$$

forces $\lambda(tb_*)^m > v$. The dominant spatial eigenvalue is positive: if it were negative, the absolutely convergent signed trace at large odd lengths would be negative, contradicting the positive physical partition function. No odd-chain gap is presumed.

For the main seed take $m = 26$, $t = 11/5$, and even reference $n = 44$. At order n , divide the residual bounds by $v^{n/m}$ to obtain

$$\begin{aligned} i &\geq (A - v)^{n/m} / v^{n/m}, \\ j &\geq \max\{2((B - v)/2)^{n/m}, (A - v)^{n/m} + 2((B - A)/2)^{n/m}\} / v^{n/m}, \\ r &\geq \mathcal{R}_{n/m}(v; D, A, B) / v^{n/m}. \end{aligned}$$

Thus i, j, r bound, respectively, the normalized I residual, J residual, and full residual at $(n, b) = (44, 539/20)$.

The same published 26-dimensional integer matrix triple is newly contracted at length 88. Put $V_{88} = \text{Tr}(T^{88})$ and $U_{88} = \text{Tr}(T^{86}O)$ as in [1, Appendix C]. The recorded integers satisfy

$$V_{88} > 0, \quad 500000U_{88} + 700741V_{88} < 0, \quad U_{88}/V_{88} \simeq -1.4014821054306783.$$

Hence $Z_{88}(b) > 1$ for all $b > 0$. The new contraction is independently reconstructed from the published entry recipe; a four-site direct wavefunction calculation checks its norm and bond-energy convention. Full integers are in `independent_results/trial88.json`; `verify_trial88.py` recomputes them using integer matrices.

If $Z_{88}(b_*) \leq M$, its largest physical Boltzmann weight exceeds one, and the remaining weights have sum less than $M - 1$. Raising them to $t \geq 1$ therefore gives

$$T(88, tb_*) \geq [1 + (M - 1)^t]^{-2}.$$

This supplies the physical defect q . No assumed physical excitation multiplicity enters this bound.

3.3 Retaining the sectors through doubling

At even (n, b) , let the positive leading eigenvalue be λ . Suppose $i \leq j \leq r$ are upper bounds for the three normalized residual masses just defined, and $1 - T(2n, b) \leq q$. Then

$$Z_n(b) \leq \lambda^n(1 + r), \quad Z_n(b, P) \leq \lambda^n(1 + j), \quad Z_n(b, C) \leq \lambda^n(1 + i).$$

Squaring each positive physical partition function bounds its value at $2b$. In units λ^{2n} the new caps are

$$\begin{aligned} D &= (1 + r)^2 = 1 + d_D, \\ B &= [(1 + r)^2 + 3(1 + j)^2]/4 = 1 + d_J, \\ A &= [(1 + r)^2 + 3(1 + j)^2 + 8(1 + i)^2]/12 = 1 + d_I. \end{aligned}$$

On the other hand, the sum of squares of the new n th-moment weights is

$$\frac{Z_{2n}(2b)}{\lambda^{4n}} = T(2n, b) \left(\frac{Z_{2n}(b)}{\lambda^{2n}} \right)^2 \geq 1 - q.$$

When $D^2/2 < 1 - q$, there is again a unique leading weight above $D/2$. For $h > 0$, let $v = 1 - h > D/2$ and define

$$R(h) = \max\{(d_D + h)^2/3, (d_I + h)^2 + (d_D - d_I)^2/3, \\ (d_I + h)^2 + (d_J - d_I)^2/2 + (d_D - d_J)^2/3, \\ (d_J + h)^2/2 + (d_D - d_J)^2/3\}.$$

This is (3) at relative order two. The strict comparison

$$q + R(h) < 2h - h^2 \tag{5}$$

forces $\lambda(2b)^n/\lambda(b)^{2n} > v$, by the same increasing-function argument as (4). At the new reference $(2n, 2b)$, we may use

$$i_+ \geq (d_I + h)^2/v^2, \\ j_+ \geq \max\{(d_J + h)^2/2, (d_I + h)^2 + (d_J - d_I)^2/2\}/v^2, \\ r_+ \geq R(h)/v^2, \quad p_+ = 1 - (1 + r_+)^{-2}, \\ q_+ \geq f(1 - (1 - q)^2(1 - p_+)), \quad f(u) = u^2/[2(1 - u)^2].$$

The r_+ estimate implies $S(2n, 2b) \geq (1 + r_+)^{-2}$. The last line follows from the original partition-function cancellation and physical squaring proof: $T(4n, b) \geq (1 - q)^2(1 - p_+)$, followed by temperature doubling. Thus only the spatial part of the update is replaced. Upward rounding of i_+, j_+, r_+, q_+ preserves all bounds.

3.4 Finite lengths, infinite tail, and verification

Each recurrence step also bounds the residual moments at the same order n at both temperatures:

$$R_b(n) \leq r, \quad R_{2b}(n) \leq s := (d_D + h)/v.$$

Choose rational $c, \rho_0, \rho_1 \in (0, 1)$ with $c^n \leq v$, $\rho_0^n \geq r$, and $\rho_1^n \geq s$. The signed trace argument gives, for every integer $n \leq A \leq L \leq B$,

$$T(L, b) \geq c^B \frac{1 - s\rho_1^{A-n}}{(1 + r\rho_0^{A-n})^2}, \tag{6}$$

whenever the numerator is positive. This is the same-reference-order interpolation: no larger-moment upper bound is used to upper-bound a smaller moment.

For shorter lengths, use (4) at $t = 2$ and $m = 18, 20, 22, 24, 26$. At every integer $L \geq m$ it gives

$$Z_L(2b_*) \geq v^{L/m} - \mathbf{1}_{L \text{ odd}} \mathcal{R}_{L/m}(v; D, A, B).$$

At even L , $M_J(L) + 3M_N(L)$ bounds the denominator partition function. At odd L one can discard the negative N eigenvalues to obtain $J_L + 3N_L \leq M_J(L) + 3N_{12}^{\text{up}} W^{L-12}$; when available, a signed polynomial upper bound on N_L is sharper. Dividing the numerator bound by the square of the denominator bound gives a purity lower bound K . The exact test $\exp(b\delta) < K/(1 - K)$, together with $K > 1/2$, proves a unique physical ground state and $\gamma_L > \delta$.

The certificate makes 237 named exact checks, in addition to its root and Bernstein-subdivision comparisons. It checks each length 32 through 82 separately at b_* , and covers the next lengths by the following intervals. The last column gives the recurrence index k , with $b_k = (539/20)2^k$.

First length	Last length	k
83	85	0
86	99	0
100	141	0
142	221	1
222	356	1
357	727	2
728	1508	3
1509	3231	4
3232	7417	5
7418	16895	6

Each adjacent pair 20, 21 through 30, 31 is checked at its respective smaller constant; combining these with the stronger bounds above it gives the stated all-integer ranges. The isolated length 18 is checked too. The machine-readable record gives every parameter and verifies that the main finite cover omits no integer. After eight sector-resolved updates the reference is $n_0 = 11264$, $b_0 = 34496/5$. The checker verifies $p_0^3 \leq w_0^2$ and $q_0 \leq w_0$, for the recorded rational w_0 . The retained asymmetric invariant (22) then applies with $w_+ = 6w^2$ and $T(L, b_k) \geq 1 - (15/2)w_k$ on $3n_k/2 \leq L \leq 3n_k$. Its dyadic intervals cover every integer $L \geq 16896$. With a rational upper enclosure $E \geq \exp((27/250)b_0)$, the final tests are

$$6Ew_0 < 1, \quad E \frac{(15/2)w_0}{1 - (15/2)w_0} < 1.$$

They prove the strict bound on the infinite tail, not just on a large finite sample. The finite checks and tail together prove Theorem 2.

The checker `verify_sector_refinement.py` uses only Python’s standard library. Root evaluations merely propose rational parameters; every direction is proved by enclosing the corresponding integer power with rational arithmetic. Positive products are rounded outward on a 10^{-500} grid. The analytic arguments are not machine-formalized. The previous five certificates and their source dependencies are retained.

Supplemental computations and limits of the search. Three additional cyclic-twist traces at b_* were computed at $n = 6, 8, 10$ using the unmodified packed-integer recurrence in [1, Appendix A], with Poisson mean 98 replacing 84. Their rational enclosures are, respectively, $.925305814 \pm 10^{-9}$, $.974160373 \pm 2 \times 10^{-8}$, and $.993475958 \pm 4 \times 10^{-7}$. For completeness, conditioning changes the contraction average by at most $2^3 3^{98}/2^{257}$; the column-rounding error is at most $257(2\sqrt{3^n} + 1)/2^{56}$. Multiplying by $\sqrt{e^{x_n} 3^n}$ gives certified norm-error upper bounds 2.1×10^{-10} , 6×10^{-9} , and 1.8×10^{-7} . The computed scaled squared norms are below one, so the trace error is at most $2D + D^2$ for a norm-error bound D . The original contraction, representative-column, and packed-lane proofs are unchanged. These traces are separately reproducible and checked, but are not used in Theorem 2. Temperature and reference-length searches, and attempts using these extra traces, are recorded in `RESEARCH_NOTES.md`. Their numerical decay rates are diagnostics, not upper bounds or optimality claims.

4 Retained version 1.4 spectral refinement

Theorem 3. *Assume the spatial-transfer representation and sector decomposition of [1, Section 3], the thermal and trial inputs used in [1, Section 5], and the periodic spectral conclusions of [2, Proposition 3.4], at the pinned revision. Then*

$$\gamma_L > \frac{7}{100} = 0.07 \quad \text{for every integer } L \geq 33, \quad J = 1,$$

and the ground state is unique on this range. Also $\gamma_{32} > 3/50$, with a unique ground state. Consequently $\gamma_L > 0.06$ for every integer $L \geq 32$, and $\liminf_{L \rightarrow \infty} \gamma_L \geq 0.07$ over all integer lengths. For positive J , multiply these lower bounds by J .

This improves the version 1.3 constant by $1/6$, and includes the additional odd length 33. All finite thermal and variational data are unchanged. The polynomial-filter and capped-mass principles below are already in [1, Section 5]; the new work is the filter coefficient selection, the sharper spectral estimates and reseeding, and their connection to the all-integer interpolation certificate.

4.1 A tighter filter and partition-function upper bounds

Put $b_* = 49/4$. The spatial spectrum consists of a J sector and three identical N sectors. Let j_k be the published center for J_k , $4 \leq k \leq 12$, with absolute error at most $\epsilon = 30 \times 10^{-6}$. These centers are $(Z_k + 3Z_k(P))/4$ from the thermal table in [1, Lemma 5.1], and are recorded in the checker. Choose

$$\begin{aligned} P(x) &= 12328 + 32348x - 499897x^2 - 339164x^3 + 1787190x^4, \\ F(x) &= x^4 P(x)^2 = \sum_{k=4}^{12} f_k x^k, \quad B_F = \sum_{k=4}^{12} (f_k j_k + \epsilon |f_k|), \\ U &= 1.0013505, \quad V = .872. \end{aligned}$$

Exact rational comparisons give $F(U), F(-U) > B_F$. Every coefficient of $P(U + y)$ and $P(-U - y)$ is positive, so both $F(U + y)$ and $F(-U - y)$ increase for $y \geq 0$. Since $F \geq 0$ on the real line and the summed filter is at most B_F , every J eigenvalue has modulus less than U . The former cap was 1.00139. We retain the published cap V in the N sectors and the published twelfth-moment bounds

$$m_J = 1.0657205, \quad m_N = .2783155.$$

For a nonnegative list of mass at most m with entries at most h , where $h < m < 2h$, the capped-mass lemma gives $\sum_i t_i^r \leq h^r + (m - h)^r$ for every real $r \geq 1$. Applying this with $t_i = |\mu_i|^{12}$ separately in each sector yields, for every even integer $k \geq 12$,

$$Z_k(b_*) \leq U^k + (m_J - U^{12})^{k/12} + 3[V^k + (m_N - V^{12})^{k/12}]. \quad (7)$$

Using the cap consistently in both the leading term and the remaining mass avoids maximizing these two terms separately. The exact certificate proves

$$Z_{32}(b_*) < B_{32} := 1.08612266, \quad Z_{120}(b_*) < B_{120} := 1.175802453. \quad (8)$$

The two cubic roots needed at $k = 32$ are enclosed by rational candidates whose directions are checked by integer powers. At $k = 120$ all exponents are integers.

4.2 Dominant-eigenvalue lower bounds and new seeds

The original argument gives a positive dominant J eigenvalue $\lambda(b_*) > .999$. Removing it and using the original twelfth-moment bounds gives the residual upper bound

$$A_{120} = (m_J - .999^{12})^{10} + 3[V^{120} + (m_N - V^{12})^{10}].$$

The trial input implies $Z_{120}(b) > 1$ at every $b > 0$. With $\ell = .999999998$, the checker verifies $\ell^{120} + A_{120} < 1$. Hence $\lambda(b_*) > \ell$. Removing this stronger leading contribution gives

$$R_{b_*}(32) < .04236466, \quad \rho_{b_*} < .872001, \quad (9)$$

where $R_b(k)$ is the residual absolute moment normalized by the positive dominant eigenvalue, and ρ_b is the largest residual absolute eigenvalue ratio. For the ratio bound, use $m_J - \ell^{12} < V^{12}$ and $V/\ell < .872001$.

At the doubled temperature, physical purity gives $Z_{32}(2b_*) \leq Z_{32}(b_*)^2 < M := B_{32}^2$. Let $\ell_2 = .9999865$ and $v = \ell_2^{32}$. The rational certificate proves

$$M/2 < v < M, \quad (M - v)^{15} < (1 - \ell_2^{120})^4. \quad (10)$$

If every $|\mu_i(2b_*)|^{32}$ were at most v , the capped-mass lemma with exponent $120/32 = 15/4$ would imply

$$1 < Z_{120}(2b_*) \leq v^{15/4} + (M - v)^{15/4} < 1,$$

a contradiction. Thus a unique eigenvalue has modulus exceeding ℓ_2 ; uniqueness follows from $2v > M$. It is positive: a unique negative dominant eigenvalue would make the signed trace negative at sufficiently large odd lengths, contradicting the positivity of the physical partition function. Absolute convergence at order 32 justifies domination of the residual at those lengths. This argument assumes no odd-chain spectral gap.

The further rational comparison

$$M < \ell_2^{32}(1 + .947852^{32})$$

gives $R_{2b_*}(32) < .947852^{32}$, and therefore $R_{2b_*}(L) < .947852^L$ for every integer $L \geq 32$. It does not give such a conclusion at smaller moment orders. In particular,

$$S(60, 2b_*) \geq (1 + .947852^{60})^{-2} > 1 - .07584.$$

For the physical seed, the shifted ground-state Boltzmann weight on 120 sites exceeds one. At b_* the sum of all other weights is less than $B_{120} - 1$; squaring them at $2b_*$ gives a normalized residual less than $(B_{120} - 1)^2$. Consequently

$$1 - S(60, 49/2) \leq .07584, \quad 1 - T(120, 49/2) \leq .059062. \quad (11)$$

Both inequalities have strict arithmetic margins.

4.3 All integer lengths and the certificate

The initial ratio satisfies

$$\frac{\lambda(2b_*)}{\lambda(b_*)^2} > \frac{.9999865}{1.0013505^2} > c_* := .997290.$$

For an integer interval $32 \leq A \leq L \leq B$, the signed traces and (9) therefore give

$$T(L, b_*) \geq \frac{c_*^B(1 - .947852^A)}{(1 + .04236466 .872001^{A-32})^2}. \quad (12)$$

At $A = B = 32$, this proves the separate claim $\gamma_{32} > .06$.

For the longer scales, insert (11) into the coupled update (17) and the signed interpolation (19) below. As in version 1.3, every recurrence output is rounded upward to 12 significant decimal digits. At the seventh doubling the certified bounds are

$$n = 7680, \quad b = 3136, \quad p = 4.34388004302 \times 10^{-71}, \quad q = 1.23795864491 \times 10^{-97}.$$

Table 1 covers every integer from 33 through 11519. For each row its purity lower bound $\mathcal{K}$ satisfies $\mathcal{K} > 1/2$ and $E_m(7b/100) < \mathcal{K}/(1 - \mathcal{K})$, using the rational exponential upper bound (35). Thus the ground state is unique and $\gamma_L > .07$ on every listed interval.

Table 1: Version 1.4 finite certificate; both parities are included.

First length	Last length	Inverse temperature b
33	34	49/4
35	48	49/4
49	99	49/4
100	153	49/2
154	208	49/2
209	387	49
388	885	98
886	2295	196
2296	5534	392
5535	11519	784

For the infinite tail take $w_0 = 2 \times 10^{-97}$. The checker verifies $p^3 \leq w_0^2$, $q \leq w_0$, and the domain of the invariant (22). Its proof below applies unchanged: $w_{k+1} = 6w_k^2$ and $T(L, 3136 \cdot 2^k) \geq 1 - (15/2)w_k$ on $11520 \cdot 2^k \leq L \leq 23040 \cdot 2^k$. With $E = E_m(3136 \times 7/100)$, the two certified inequalities

$$E 6w_0 < 1, \quad E \frac{(15/2)w_0}{1 - (15/2)w_0} < 1$$

prove $\gamma_L > .07$ on this entire infinite tail by the argument following (23). The intervals meet without a missing integer, proving Theorem 3.

The checker `verify_spectral_refinement.py` performs 111 exact checks, including the filter, seed estimates, ten finite intervals, the length-32 consequence, and the infinite tail. It imports the directed rational arithmetic helpers of the retained version 1.3 checker. Numerical optimization selected the candidate filter only; the certificate contains fixed integer coefficients and requires no optimizer or floating-point acceptance decision.

5 Retained version 1.3 length-adaptive bound

Theorem 4. *Assume the signed spatial-transfer representation of [1, Proposition 3.1] and the periodic spectral estimates of [2, Proposition 3.4 and its proof], including their finite-input premises. Then*

$$\gamma_L > \frac{3}{50} = 0.06 \quad \text{for every integer } L \geq 34,$$

and the ground state is unique throughout this range. In particular, every odd $L \geq 35$ is included, and $\liminf_{L \rightarrow \infty} \gamma_L \geq 3/50$ over all integer lengths. For a positive exchange coupling J , multiply the bound by J .

5.1 Inputs and interpolation

Write $\lambda(b) > 0$ for the unique eigenvalue of largest modulus of the spatial operator, and define its residual absolute moment by

$$R_b(m) = \sum_{i \neq 0} \left| \frac{\mu_i(b)}{\lambda(b)} \right|^m.$$

All sums used here converge absolutely. Let $b_* = 49/4$. The proof of the cited companion proposition supplies

$$R_{b_*}(32) < .0442, \quad \rho_{b_*} < .874, \quad R_{2b_*}(32) < .9525^{32}, \quad (13)$$

$$\lambda(b_*) \leq 1.00139, \quad \lambda(2b_*)^{72} > .968, \quad (14)$$

$$S(60, 49/2) > .9002, \quad T(120, 49/2) > .9372. \quad (15)$$

Here ρ_b is the largest nondominant absolute eigenvalue ratio. These are periodic estimates, without boundary fields. The reseeding comparisons were checked by the version 1.1 certificate retained below. In particular, the rational number $c_* = .9967754$ satisfies

$$(c_* 1.00139^2)^{72} < .968, \quad \frac{\lambda(2b_*)}{\lambda(b_*)^2} > c_*.$$

Increasing moment orders in (13) gives, for every integer $L \geq 32$,

$$T(L, b_*) \geq \frac{c_*^L (1 - .9525^L)}{(1 + .0442 .874^{L-32})^2}. \quad (16)$$

The first residual bound uses the ratio ρ_{b_*} term by term. For the second, the sequence norm inequality gives $R_{2b_*}(L) \leq R_{2b_*}(32)^{L/32} < .9525^L$.

For the subsequent scales, suppose n is even and $1 - S(n, b) \leq p$, $1 - T(2n, b) \leq q$. With $f(z) = z^2/[2(1 - z)^2]$, put

$$a = 1 - (1 - p)^2(1 - q), \quad p_+ \geq f(a), \quad q_+ \geq f(1 - (1 - q)^2(1 - p_+)). \quad (17)$$

The coupled update, proved in Lemma 4 below, bounds the defects at the next scale $(2n, 2b)$ by p_+, q_+ . Cancellation also gives $S(n, 2b) \geq 1 - a$ at the *same* reference length. When $p, a < 1/2$, the concentration lemma below yields

$$R_b(n) \leq d(p), \quad R_{2b}(n) \leq d(a), \quad d(z) = \frac{z}{2 - 3z}.$$

The leading spatial weight at $(2n, 2b)$ is at least $1 - p_+$. Since $Z_{2n}(b) \geq \lambda(b)^{2n}$, it follows that

$$\left(\frac{\lambda(2b)}{\lambda(b)^2} \right)^{2n} \geq (1 - p_+) \frac{Z_{2n}(2b)}{Z_{2n}(b)^2} \geq (1 - p_+)(1 - q) =: g. \quad (18)$$

Choose rational numbers $0 < c, \rho_0, \rho_1 < 1$ with $c^{2n} \leq g$, $\rho_0^n \geq d(p)$, and $\rho_1^n \geq d(a)$. Each residual eigenvalue ratio is bounded by the corresponding ρ . For any integer interval $n \leq A \leq L \leq B$, the signed trace therefore gives

$$T(L, b) \geq \frac{c^B (1 - d(a) \rho_1^{A-n})}{(1 + d(p) \rho_0^{A-n})^2} =: \mathcal{K}(A, B; b), \quad (19)$$

provided the displayed numerator factor is positive. No parity restriction is imposed on L . The numerator uses the lower bound on the signed residual, and the denominator uses its upper bound.

5.2 Finite coverage

Starting with $n = 60$, $b = 49/2$, $p = .0998$, and $q = .0628$, the checker iterates (17), rounding each output upward to 12 significant decimal digits. The rounded p_+ is used in the second update. At the seventh doubling it obtains

$$n = 7680, \quad b = 3136, \quad p = 2.30218227586 \times 10^{-56}, \quad q = 1.75070874875 \times 10^{-83}. \quad (20)$$

All these decimals are exact upper bounds. The full sequence and rational interpolation parameters are in the machine-readable certificate.

For each interval in Table 2, the checker uses (16) or (19) to certify

$$\mathcal{K} > \frac{1}{2}, \quad E_m(3b/50) < \frac{\mathcal{K}}{1 - \mathcal{K}}, \quad (21)$$

where E_m is the rational exponential upper bound in (35). For the initial four intervals, the lower bound uses c_*^B , $.9525^A$, and $.0442.874^{A-32}$. The largest physical Gibbs weight exceeds $1/2$, so the ground state is unique; the ratio of a next-energy weight to it is at most $(1 - \mathcal{K})/\mathcal{K}$. Thus (21) proves $\gamma_L > 3/50$ for every integer $34 \leq L \leq 11519$.

Table 2: Certified finite intervals; every integer in each row is covered.

First length	Last length	Inverse temperature b
34	35	49/4
36	46	49/4
47	84	49/4
85	116	49/4
117	164	49/2
165	202	49/2
203	302	49
303	421	49
422	814	98
815	1937	196
1938	4748	392
4749	11519	784

The root calculations in parameter selection are proposals only. Every root direction is certified by an integer-power comparison over the rationals. Positive products in those comparisons and in (19) are rounded outward to multiples of 10^{-140} . Floor or ceiling of the integer quotient gives the corresponding enclosure; induction through exponentiation by squaring preserves its direction. The chosen interval covering is then checked for both its inequalities and the absence of missing integers.

5.3 An asymmetric invariant for the infinite tail

Suppose

$$p^{3/2} \leq w, \quad q \leq w, \quad 0 < w \leq 10^{-6}. \quad (22)$$

Write $x = w^{1/3} \leq 1/100$, so $p \leq x^2$ and $q \leq x^3$. Since $a \leq 2p + q \leq (2 + x)x^2$, the exact scalar comparisons

$$\frac{(2.01)^2}{2(1 - 2.01 \times 10^{-4})^2} < 3, \quad \frac{(2.03)^2}{2(1 - 2.03 \times 10^{-6})^2} < 3$$

imply, using the unrounded upper updates,

$$p_+ \leq 3x^4, \quad q_+ \leq f(2x^3 + 3x^4) \leq 3w^2.$$

Consequently $p_+^{3/2} \leq \sqrt{27} w^2 < 6w^2$ and $q_+ \leq 6w^2$. The invariant propagates with $w_+ = 6w^2$. For every integer $3n/2 \leq L \leq 3n$, increasing the absolute-moment order gives

$$R_b(L) \leq d(p)^{3/2} \leq w, \quad R_{2b}(L) \leq d(a)^{3/2} \leq (2.01)^{3/2} w < 3w.$$

We used $d(z) \leq z$ for $0 \leq z \leq 1/3$; both p and a lie in this domain. Also $g \geq 1 - q - p_+ \geq 1 - 1.03w$. By (18), the leading-eigenvalue contribution at length L is at least $g^{L/(2n)} \geq g^2 \geq 1 - 2.06w$. It follows that

$$T(L, b) \geq \frac{(1 - 2.06w)(1 - 3w)}{(1 + w)^2} \geq 1 - 7.06w \geq 1 - \frac{15}{2}w > \frac{1}{2}. \quad (23)$$

Here $(1 + w)^{-2} \geq 1 - 2w$, and multiplication of the three nonnegative factors bounds their defect by the sum of the individual defects.

At (20), take $w_0 = 2 \times 10^{-83}$. The checker verifies $p^3 \leq w_0^2$, $q \leq w_0$, and $6w_0 < 1$. Let

$$n_k = 7680 \cdot 2^k, \quad b_k = 3136 \cdot 2^k, \quad w_k = \frac{1}{6}(6w_0)^{2^k}, \quad B_0 = 15/2.$$

The invariant and (23) apply on every interval $[11520 \cdot 2^k, 23040 \cdot 2^k]$. On each interval the ground state is unique, and

$$e^{-b_k \gamma_L} \leq \frac{B_0 w_k}{1 - B_0 w_k} \leq \frac{B_0}{6(1 - B_0 w_0)} (6w_0)^{2^k}.$$

Set $E = E_m(3136 \times 3/50) \geq e^{3136(3/50)}$. The final two exact exponential checks are

$$E \cdot 6w_0 < 1, \quad E \frac{B_0 w_0}{1 - B_0 w_0} < 1.$$

Therefore

$$e^{b_k(3/50)} e^{-b_k \gamma_L} \leq \frac{B_0}{6(1 - B_0 w_0)} (E \cdot 6w_0)^{2^k} < 1.$$

This proves the strict target bound for every integer $L \geq 11520$. Together with the finite coverage, it proves Theorem 4.

The certificate `verify_adaptive_bound.py` implements 88 exact checks. Its analytic assumptions and the proof above remain outside machine-formal verification. The gain comes from retaining unequal defects, changing the interpolation intervals, and using an invariant matched to their different decay rates. No extra finite input is used.

6 Including odd lengths

This section retains the version 1.2 argument and its smaller constant.

Theorem 5. *Assume the signed spatial-moment representation of [1, Proposition 3.1] and the periodic concentration estimates, positive dominant eigenvalues, and adjacent-temperature eigenvalue ratios of [2, Proposition 3.4], including their finite-input premises. The periodic chain has a unique ground state and*

$$\gamma_L > \frac{\log(125/39)}{49} > 0.02377 \quad \text{for every integer } L \geq 120.$$

In particular, this holds for every odd $L \geq 121$, and $\liminf_{L \rightarrow \infty} \gamma_L \geq \log(125/39)/49$ over all integer lengths. For a positive exchange coupling J , multiply the bounds by J .

We first record the concentration bound on the domain needed below. The companion states its rational version for defects at most .1; its elementary proof extends to all defects below 1/2.

Lemma 3 (Concentration). *Let (p_i) be a finite or countable probability list with $\sum_i p_i^2 \geq 1 - z$, where $0 \leq z < 1/2$. Its largest weight p_0 is unique, and*

$$\frac{\sum_{i \neq 0} p_i}{p_0} \leq d(z), \quad d(z) = \frac{z}{2 - 3z}.$$

Proof. A summable nonzero nonnegative list attains its maximum. Since $\sum_i p_i^2 \leq p_0$, one has $p_0 \geq 1 - z > 1/2$. Put $x = 1 - p_0 \leq z$. Then $2x(1 - x) \leq 1 - \sum_i p_i^2 \leq z$, so $x \leq z/[2(1 - x)] \leq z/[2(1 - z)]$. Applying the increasing function $x \mapsto x/(1 - x)$ gives the result; the displayed upper bound for x is below one when $z < 1/2$. $\square$

Proof of Theorem 5. Reindex the companion's scales $j \geq 2$ by $k = j - 2 \geq 0$, and set

$$n_k = 120 \cdot 2^k, \quad \beta_k = 49 \cdot 2^k, \quad u_k = \frac{1}{6} \left(\frac{39}{125} \right)^{2^k}. \quad (24)$$

Thus $u_0 = 13/250$ and $u_{k+1} = 6u_k^2 \leq u_k$. Write $n = n_k$, $\beta = \beta_k$, $u = u_k$, and $v = u_{k+1}$. The cited propositions supply real spatial eigenvalues $\mu_i(b)$ with absolutely convergent moments and

$$Z_L(b) = \sum_i \mu_i(b)^L \quad (\text{every integer } L \geq 4).$$

At $b = \beta, 2\beta$, their respective unique eigenvalues of maximal modulus are positive; denote these by λ, λ' . With $c = \lambda'/\lambda^2$, the companion gives

$$S(n, \beta) \geq 1 - u, \quad T(2n, \beta) \geq 1 - u, \quad 0 < c \leq 1, \quad c^{2n} \geq (1 - u)(1 - v). \quad (25)$$

Positivity of the leading eigenvalue does not assume an odd-chain gap: the companion proves it from positivity of physical partition functions and dominance in the signed trace at arbitrarily large odd lengths.

The necessary control at the *same* reference length n and the next temperature follows by partition-function cancellation:

$$S(n, 2\beta) = \frac{S(n, \beta)^2 T(2n, \beta)}{T(n, \beta)^2} \geq (1 - u)^3 \geq 1 - 3u. \quad (26)$$

Here $T(n, \beta) \leq 1$. Because n is even, the two spatial lists at this length are probability lists. Lemma 3, applicable since $3u \leq 39/250 < 1/2$, gives

$$\sum_{i \neq 0} \left| \frac{\mu_i(\beta)}{\lambda} \right|^n \leq d(u), \quad \sum_{i \neq 0} \left| \frac{\mu_i(2\beta)}{\lambda'} \right|^n \leq d(3u). \quad (27)$$

Each absolute eigenvalue ratio is at most one. Increasing its exponent from n to any integer $L \geq n$ therefore decreases every summand. Absolute convergence and the triangle inequality yield

$$Z_L(\beta) = \lambda^L (1 + \epsilon_L), \quad |\epsilon_L| \leq d(u), \quad Z_L(2\beta) = (\lambda')^L (1 + \epsilon'_L), \quad |\epsilon'_L| \leq d(3u).$$

These estimates allow negative nondominant eigenvalues. No probability interpretation of an odd spatial moment is used. In particular, (26) is essential: a higher-moment upper bound alone cannot bound a lower moment by taking a root.

For every integer $L \in [n, 2n]$, $c^L \geq c^{2n}$, and hence

$$T(L, \beta) \geq K(u) := \frac{(1 - u)(1 - 6u^2)(1 - d(3u))}{(1 + d(u))^2} \geq 1 - \frac{9}{2}u. \quad (28)$$

To prove the last inequality on the entire interval $0 \leq u \leq 13/250$, use the exact identity

$$K(u) - \left(1 - \frac{9}{2}u\right) = \frac{u[4 - 60u + 243u^2(1 - 2u) + 324u^4]}{2(2 - 9u)(1 - u)} \geq 0. \quad (29)$$

Indeed, $4 - 60u \geq 22/25 > 0$, $1 - 2u > 0$, and both denominator factors are positive on that interval. In particular, $T(L, \beta) \geq 383/500 > 1/2$.

Let w_0 be the largest physical Gibbs weight, counting eigenvectors with multiplicity. Since $w_0 \geq T(L, \beta) > 1/2$, the ground state is unique. For an eigenvector of the next distinct energy,

$$\begin{aligned} e^{-\beta_k \gamma_L} &= \frac{w_1}{w_0} \leq \frac{(9/2)u_k}{1 - (9/2)u_k} \\ &\leq \frac{375}{383} \left(\frac{39}{125}\right)^{2^k} < \left(\frac{39}{125}\right)^{2^k}. \end{aligned}$$

Taking logarithms and dividing by $\beta_k = 49 \cdot 2^k$ proves the claimed strict bound on each interval $[120 \cdot 2^k, 240 \cdot 2^k]$. These intervals cover all integer $L \geq 120$. The rounded consequence follows from the exact exponential comparison described in (35). The liminf inequality follows from the uniform finite-length bound. $\square$

6.1 Certificate and contribution

The script `verify_signed_extension.py` makes 21 exact checks: the polynomial coefficient identity in (29), endpoint inequalities proving its sign over the full interval, the purity floor, the strict prefactor $375/383 < 1$, and the comparison with .02377. It uses no sampling grid or numerical logarithm in a decision. The concentration and signed-trace steps above are analytic proofs, not machine-formalized theorems.

This version adds the signed-moment interpolation argument to the even-length corollary retained below. It uses more of the companion's already proved spectral information and requires no new thermal or variational input. The argument is conditional on those results; it is not an independent proof of the original Haldane-gap claim. No exhaustive priority search for this parity extension has been made.

7 The retained version 1.1 corollary

Despite its boundary-field title, Section 3 of [2] uses the periodic partition function $Z_n(b)$ of precisely the Hamiltonian above, without endpoint fields. Its separate boundary partition functions do not enter the following argument.

Theorem 6. *Assume the spatial-moment representation of [1, Proposition 3.1] and the periodic concentration estimates of [2, Proposition 3.4], including their finite-input premises. The periodic chain has a unique ground state and*

$$\boxed{\gamma_L > \frac{\log(125/39)}{49} > 0.02377 \quad \text{for every even } L \geq 120.}$$

Consequently $\liminf_{L \rightarrow \infty, L \text{ even}} \gamma_L \geq \log(125/39)/49$. For a positive exchange coupling J , multiply these bounds by J .

Proof. The companion sets $b_j = 2^j(49/4)$ and, for $j \geq 2$, $n_j = 120 \cdot 2^{j-2}$, $M_j = 2n_j$, and $u_2 = .052$. Its Proposition 3.4 gives $1 - S(n_j, b_j) \leq u_j$ and $1 - T(M_j, b_j) \leq u_j$. In particular, $j = 2$ gives

$$1 - S(120, 49) \leq \frac{13}{250}, \quad 1 - T(240, 49) \leq \frac{13}{250}. \quad (30)$$

Apply [1, Proposition 4.3] with the new starting parameters

$$n_0 = 120, \quad \beta_0 = 49, \quad u_0 = \frac{13}{250}, \quad C = 6, \quad r = Cu_0 = \frac{39}{125}.$$

Here β_0 is the companion's b_2 , not its b_0 . Writing $G(u) = \frac{1}{2}(\sum_{k=1}^3(1-u)^{-k})^2$, exact substitution gives

$$0 < u_0 < \frac{1}{6}, \quad G(u_0) = \frac{989224080031250}{177210755074809} < 6, \quad r < 1, \quad C(1 - 3u_0) = \frac{633}{125} > 3.$$

The two seed assumptions and every scalar hypothesis of that proposition therefore hold. Its interpolation step covers all even $L \geq n_0$, and its conclusion is uniqueness and $\gamma_L > \log(1/r)/\beta_0 = \log(125/39)/49$. The strict comparison with .02377 follows from the rational Taylor upper bound in (35), which verifies $e^{49(2377/100000)} < 125/39$. $\square$

7.1 Scalar certificate and attribution

The script `verify_companion_corollary.py` checks the substitution above and the reseeding comparisons in the proof of [2, Proposition 3.4]. That proof starts from the finite periodic inputs of [1, Corollary 5.6], including the 72- and 120-site trial bounds already covered by our recorded finite verification. It requires no new boundary-field thermal calculation for the periodic seed used here.

In the companion's notation, the checker verifies the rational majorants $D(32) < .0442$ and $D(120) < .00000025$, the residual spectral ratio bound, and the seeds $S(60, 49/2) > .9002$ and $T(120, 49/2) > .9372$. The coupled update then gives defects below .0502 and .0198, both below .052 at the scales in (30). Cube-root bounds are checked by cubing positive rational majorants; no floating-point root or logarithm determines a pass/fail decision.

The underlying reseeding argument and uniform-gap criterion are both OpenAI's published results. Our contribution here is their explicit combination, its scalar certificate, and the resulting comparison. The value $\log(125/39)/49 = 0.023770450840258259\dots$ was extracted during our prior-art review; we have not located it explicitly stated in the examined sources. This limited observation does not establish priority. Version 1.0's comparison omitted this consequence of the companion paper; the present version corrects that omission.

8 The retained version 1.0 refinement

The earlier, weaker result remains a separate reproducible extension of the periodic paper's final seed pair. Its dependencies on [1] are the following two inputs:

1. For each $b > 0$, a real finite or countable sequence $\lambda_i(b)$ obeys $\sum_i |\lambda_i(b)|^4 < \infty$ and $Z_L(b) = \sum_i |\lambda_i(b)|^L$ for every even $L \geq 4$ (Proposition 3.1).
2. The initial estimates are

$$1 - S(2304, 784) \leq \frac{84513}{10^7}, \quad 1 - T(4608, 784) \leq \frac{27493}{5 \cdot 10^6} \quad (31)$$

(Proposition 5.5, including its finite certificates).

All references are to the fixed upstream revision listed at the end.

Theorem 7. *Under these inputs, the ground state is unique and*

$$\gamma_L > \frac{47}{10000} \quad \text{for every even } L \geq 2304.$$

Consequently $\liminf_{L \rightarrow \infty, L \text{ even}} \gamma_L \geq 47/10000$. For any positive exchange coupling J , multiply these bounds by J .

The argument below rederives the needed bootstrap steps of [1, Lemma 4.2 and Proposition 4.3]. It does not establish the two inputs independently of the upstream analytic argument.

9 Purity estimates and gap extraction

For an even reference length $n \geq 4$, both quantities in (1) are purities of probability distributions, so they belong to $(0, 1]$. For spatial purity use $|\lambda_i(b)|^n/Z_n(b)$; for physical purity use the Gibbs weights, counting each eigenvector with multiplicity. If $P = \sum_i w_i^2$, the normalized squares have purity P' satisfying

$$1 - P' = \frac{2 \sum_{i < k} w_i^2 w_k^2}{P^2} \leq \frac{2(\sum_{i < k} w_i w_k)^2}{P^2} = f(1 - P), \quad f(x) = \frac{x^2}{2(1 - x)^2}.$$

Here f is increasing on $[0, 1]$.

Lemma 4 (Rounded update). *For even $n \geq 4$ and $b > 0$, if $1 - S(n, b) \leq p$ and $1 - T(2n, b) \leq q$, with $p, q \in [0, 1]$, any $p_+, q_+ \in [0, 1]$ satisfying*

$$p_+ \geq f(1 - (1 - p)^2(1 - q)), \quad q_+ \geq f(1 - (1 - q)^2(1 - p_+)) \quad (32)$$

bound $1 - S(2n, 2b)$ and $1 - T(4n, 2b)$ respectively.

Proof. Cancellation of partition functions gives

$$S(n, 2b) = \frac{S(n, b)^2 T(2n, b)}{T(n, b)^2}, \quad T(4n, b) = \frac{T(2n, b)^2 S(2n, 2b)}{S(2n, b)^2}.$$

The denominators are at most one. Apply the probability-squaring inequality first in the spatial variable and then in the temperature variable. Use the already rounded p_+ in the second application. $\square$

Lemma 5 (Intermediate lengths). *Under the same hypotheses, put $K = (1 - p)^2(1 - q)$. If $K > 1/2$, every even $L \in [n, 2n]$ has a unique ground state and*

$$\gamma_L \geq \frac{1}{b} \log \frac{K}{1 - K}. \quad (33)$$

Proof. The monotonicity of sequence ℓ^r norms, applied to the spatial moments at b and $2b$, yields

$$Z_L(b) \leq Z_n(b)^{L/n}, \quad Z_L(2b) \geq Z_{2n}(2b)^{L/(2n)}.$$

Hence

$$T(L, b) \geq [T(2n, b) S(n, b)^2]^{L/(2n)} \geq K.$$

The last step uses a base in $(0, 1]$ and an exponent at most one. The largest Gibbs weight w_0 is at least the purity, because $\sum_i w_i^2 \leq (\max_i w_i) \sum_i w_i$. Thus $w_0 > 1/2$, excluding ground-state degeneracy. For one eigenvector at the next energy,

$$e^{-b\gamma_L} = \frac{w_1}{w_0} \leq \frac{1 - w_0}{w_0} \leq \frac{1 - K}{K}.$$

Taking logarithms proves the assertion. $\square$

Table 3: Upward-rounded purity-defect bounds.

j	n_j	b_j	p_j	q_j
0	2304	784	0.0084513	0.0054986
1	4608	1568	2.59×10^{-4}	6.45×10^{-5}
2	9216	3136	1.70×10^{-7}	8.35×10^{-9}
3	18432	6272	6.07×10^{-14}	1.40×10^{-16}
4	36864	12544	7.39×10^{-27}	3.93×10^{-32}
5	73728	25088	1.10×10^{-52}	3.09×10^{-63}

10 A finite certificate and the infinite tail

Use the exact rational numbers in Table 3 with $n_j = 2304 \cdot 2^j$ and $b_j = 784 \cdot 2^j$. Every displayed decimal denotes an exact rational, not a floating-point approximation. The script `verify_gap_improvement.py` checks (32) at each of the five steps.

For $j = 0, \dots, 4$, set $K_j = (1 - p_j)^2(1 - q_j)$. The certificate verifies

$$K_j > \frac{1}{2}, \quad e^{(47/10000)b_j} < \frac{K_j}{1 - K_j}. \quad (34)$$

Lemma 5 therefore proves the strict target bound throughout $2304 \leq L \leq 73728$, restricted to even lengths.

For clarity, the exponential comparisons use no numerical transcendental evaluation. Given rational $x \geq 0$, choose $m = 4\lfloor x \rfloor + 64$. Since $\rho = x/(m+2) < 1$, the Taylor series and the ratio bound for its remaining terms imply

$$e^x \leq E_m(x) := \sum_{k=0}^m \frac{x^k}{k!} + \frac{x^{m+1}}{(m+1)!} \frac{1}{1 - x/(m+2)}. \quad (35)$$

Every term on the right is rational. The script verifies the stronger inequality obtained by replacing e^x with $E_m(x)$ in (34).

The final row controls all remaining lengths. Let

$$u = 1.10 \times 10^{-52}, \quad C = 5, \quad r = Cu, \quad G(v) = \frac{1}{2}((1-v)^{-1} + (1-v)^{-2} + (1-v)^{-3})^2.$$

Exact checks give $0 < u < 1/6$, $G(u) < C$, $r < 1$, and $C(1 - 3u) > 3$. Define $u_k = C^{-1}r^{2^k}$ for $k \geq 0$. Both defects at $(2^k n_5, 2^k b_5)$ are bounded by u_k : indeed, $u_{k+1} = Cu_k^2 < u_k$, and

$$f(1 - (1 - u_k)^3) = G(u_k)u_k^2 \leq Cu_k^2.$$

Lemma 4 then applies to both coordinates by induction. For even $L \in [2^k n_5, 2^{k+1} n_5]$, the proof of Lemma 5 gives $w_0 \geq (1 - u_k)^3 \geq 1 - 3u_k$. Consequently

$$e^{-(2^k b_5) \gamma_L} \leq \frac{3u_k}{1 - 3u_k} \leq \frac{3}{C(1 - 3u)} r^{2^k} < r^{2^k}.$$

These intervals cover every even $L \geq n_5$, proving

$$\gamma_L > \frac{\log(1/(Cu))}{b_5} = \frac{\log(10^{53}/55)}{25088} = 0.004704626783460297 \dots > \frac{47}{10000}. \quad (36)$$

The final strict comparison is certified by $E_m((47/10000)25088) < 10^{53}/55$, again over the rationals. Together with the five finite intervals this proves Theorem 7. A strict bound for each finite length yields the stated non-strict bound on the liminf.

11 Reproducibility and scope

The version 1.0 quantitative gain comes from retaining the unequal seed bounds and performing a few more coupled updates before using a common envelope. The upstream theorem instead rounds both seed defects to $1/100$ and uses $C = 5$, giving $\log(20)/784$. No new thermal or variational input is required for this refinement. This is a comparison with that paper’s stated constant, not with every consequence of the published pair of papers. Theorem 6 is stronger in both constant and length range. We make no priority or optimality claim.

Theorem 5 further removes the parity restriction without changing the version 1.1 constant. Theorem 4 improves both the constant and the starting length using the same finite inputs. Theorem 3 then sharpens the spectral inputs and raises the bound again, without changing the finite computation data. Theorem 2 adds the sector-resolved recurrence, simultaneous moment bounds, and the newly computed 88-site trial. Theorem 1 adds the complete cyclic table, geometric rounding bounds, and tight purity extraction. The accompanying repository contains all seven gap certificates, the upstream source and verification material at the fixed commit, fresh verification results, and `independent_audit.py`. The latter reconstructs selected computations from the printed definitions and tables without importing upstream verifier functions. The recorded audit, completed on 7 October 2026, comprises:

- A full fresh run of the upstream verifier, with all checks passing and all finite results newly recomputed. It reproduces 27 thermal certificates, the 72- and 120-site variational contractions, and the associated arithmetic and scalar checks.
- A separate arbitrary-precision integer reconstruction of both variational contractions. The complete numerator and norm integers match the upstream evidence. A direct four-site wavefunction calculation also agrees with the contraction formula.
- A separate full-spectrum diagonalization on chains of four to eight sites, including twists, in 16 cases. All values lie within the printed tolerances. This floating-point calculation is a diagnostic; it does not provide rigorous thermal enclosures.

The full upstream run took approximately 22 minutes with six requested workers, Python 3.12.14, NumPy 2.3.5, and SciPy 1.17.0. All seven bound certificates need only Python’s standard library. Run `make check` with assertions enabled to check them and the local energy and both cyclic-trace enclosures. For the current main certificate, run

```
python3 verify_resolved_refinement.py
```

All nine new cyclic totals were recomputed from the distributed evaluator. To repeat this calculation and its small independent cross-checks, run

```
make extended-cyclic
```

This needs a C++17 compiler with OpenMP and NumPy/SciPy for the odd-chain check. The recorded eight-CPU run took about 6.5 minutes. The retained 88-site trial can be reconstructed with `make trial88`. These new computations do not constitute a fresh complete upstream run. Rational comparisons determine all numerical pass/fail decisions; decimal logarithms and floating-point margins are display-only. The companion’s selected analytic sources are preserved unmodified under `source_boundary/`, with Git blob identities recorded in `boundary_source_manifest.json`.

Neither successful execution nor this note is a formal verification of the upstream analytic assumptions. No claim is made here for the same numerical bound 0.145 at lengths below 24, arbitrary open boundaries, or uniqueness of all infinite-volume ground states. The constants are conditional lower bounds, not estimates of the actual physical gap.

Autonomous authorship and human contribution. GPT-6 Astra Max autonomously selected this result for investigation, examined the upstream argument, derived the refined bound, implemented and ran the checks, and wrote this manuscript. During the user-requested prior-art review, the agent identified the stronger corollary, verified its parameter substitution, and prepared version 1.1. When the user asked about odd lengths, the agent derived the signed-moment extension, implemented its certificate, and wrote version 1.2. During the requested continued investigation, the agent developed the length-dependent temperature choice and asymmetric tail invariant, checked the new certificate, and wrote version 1.3. Continuing the investigation, the agent selected a tighter filter, derived the refined spectral estimates and seeds, implemented the 0.07 certificate, and wrote version 1.4. In version 1.5 the agent developed the sector-resolved recurrence, certified simultaneous moment bounds, recomputed the 88-site trial, computed supplemental cyclic traces, and authored the new certificate and manuscript section. In version 1.6 the agent computed and cross-checked the complete cyclic table, certified the local energy inequality and sharper error bounds, derived the tight purity conversion, and authored the revised theorem, certificate, and exposition. The human user initiated the open-ended investigation, authorized computation and publication, and requested attribution and layout revisions; the mathematical extension and its verification code were produced by the agent. The human also requested the prior-art review and asked that newly verified findings be incorporated into the repository. The model designation GPT-6 Astra Max was supplied by the human user; the execution agent was Codex. A backend model snapshot identifier was not available in the session. The upstream mathematical construction, reseeding argument, and input certificates are credited to [1, 2]. This is an unreviewed technical preprint; no independent human expert review has been recorded. Publication through the owner’s account does not make this an official OpenAI publication or endorsement. Corrections are welcome through the repository’s issue tracker.

References

- [1] OpenAI, *The periodic spin-one Haldane gap*, 24 September 2026, OpenAI Math Release preprint. Repository <https://github.com/openai/math>, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. File `preprints/The-periodic-spin-one-Haldane-gap-September-24-2026/paper.pdf`. The companion analytic source and finite verifier are at the same revision.
- [2] OpenAI, *A boundary-field gap for the spin-one Heisenberg chain*, 24 September 2026, OpenAI Math Release preprint. Repository <https://github.com/openai/math>, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. File `preprints/A-boundary-field-gap-for-the-spin-one-Heisenberg-chain-September-24-2026/paper.pdf`. Section 2 fixes the periodic normalization; Proposition 3.4 supplies the periodic concentration estimates used here.